

Vityarthi
Fundamentals of AI &
ML Project Report

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Project Report: Smart Attendance Risk Predictor

Introduction

For this project, I chose a very common but important problem faced by college students which is low attendance. In many colleges, attendance is directly linked to eligibility for exams, internal marks, and overall academic standing. A student may not always realize when their attendance is slowly dropping below the safe limit, and by the time they notice, it already gets too late. I wanted to build something decent, practical, and relevant to student life. That is how I came up with the idea of a **Smart Attendance Risk Predictor** using the basic concepts learned in the course of Fundamentals of AI & ML.

The goal of this project was not to create a highly complex system, but to build a meaningful one that could give a clear warning based on attendance data. I designed the project to take the number of classes held and attended as an input, calculate attendance percentage, and predict whether the student is safe or at risk based on the 75% attendance rule. This made the project useful, easy to understand, and closely connected to a real-world academic issue.

Problem Chosen

The problem I focused on was the lack of early awareness among students regarding attendance shortage. Many students keep track of classes casually and only check their attendance when there is already a serious issue. In such cases, they may discover that they are below the required limit and cannot fix the problem easily. This can lead to stress, loss of marks, or even being debarred from exams.

I chose this problem because it is realistic and relatable. Almost every college student has faced the pressure of maintaining attendance at some point. It is also a problem that can be solved in a simple and useful way with AI/ML concepts. Since attendance is measurable and numerical, it was a good fit for a machine learning project.

Why This Problem Matters

This problem matters because attendance affects both academic progress and student confidence. When attendance drops, students often feel anxious and unsure about whether

they can recover it in time. A small tool that predicts attendance risk early can help students become more responsible and informed.

It also matters because students do not always calculate attendance regularly. A simple prediction system can save time. Instead of manually counting every time, a student can enter their details and instantly know whether they are safe in terms of attendance. This kind of project is practical because it connects technology with a daily student concern. It shows how AI and ML can be used for something useful, not just for large or advanced applications.

Approach Used

I built the project using a very simple machine learning approach. First, I created a small dataset with two main inputs: **classes held** and **classes attended**. From these values, I calculated the attendance percentage and the number of missed classes. These values were used to train the model.

I used a basic classification model to decide whether a student was **safe** or **at risk**. The decision rule was based on the 75% attendance requirement. If the attendance percentage was 75% or more, the student was marked safe. If it was below 75%, the student was marked at risk. The model accepts user input, so a student can enter their own number of classes held and attended, and the system will calculate the percentage and show the result.

Key Decisions I Made

One of the most important decisions I made was to use the **75% rule** exactly as it is commonly followed in colleges. I made sure that if the attendance is exactly 75%, the student is considered safe. That detail was important because it makes the logic more accurate and realistic.

I also decided to show the attendance percentage in the output. This was useful because it gives the student a clear explanation of the result rather than just saying “safe” or “at risk.” It makes the program more transparent and practical.

Challenges Faced

One challenge was deciding whether to make the project fully machine-learning based or partly rule-based. Since attendance safety depends on a fixed college rule, a pure ML model was not enough by itself. I solved this by combining both approaches, that are, the model predicts based on the data, while the final decision still respects the 75% rule. Another challenge was keeping the code simple while still making it meaningful. A small technical challenge was handling the input and output in a clean way without using extra libraries.

What I Learned

This project taught me that AI and ML do not always need to be complicated to be useful. Even a simple tool can solve a real problem if it is designed thoughtfully. I learned how to identify a problem from everyday student life and turn it into a practical solution.

I also learned the importance of clarity in project design. A good project is not just about writing code, it is also about choosing the right problem, making sensible decisions, and explaining the solution clearly. Working on this project improved my understanding of classification, feature preparation, user input handling, and rule-based decision making.

Most importantly, I learned that machine learning projects should connect to real needs. The attendance predictor may be small, but it solves something students actually face.

Conclusion

The Smart Attendance Risk Predictor was a simple but useful project built around a real student problem. It helped me apply the basics of AI and ML in a practical way while keeping the solution easy to understand. By focusing on attendance risk, I created a tool that can give students early awareness and help them stay within the safe limit. The project also strengthened my confidence in planning, coding, and explaining an ML-based solution. Overall, it was a valuable learning experience that showed me how even a small project can have real purpose when it is built around a genuine problem.